Week 8 Notes

Limits and Continuity:
Limits, Derivatives, L'Hôpital's Rule, Tangents and Linear Approximation

Course: Mathematics–1

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Lecture 1 — Limits and Continuity

Recalling one-sided limits; continuity at a point; rules for limits; the sandwich principle; standard limits

1 Recalling limits of a function

In the previous lectures we met the notion of a limit — first for sequences, then for functions. Let us quickly recall the three definitions before building on them.

One-sided and two-sided limits

For a function f and a point a:

$$\lim_{x \to a^-} f(x) = L$$

means that as x approaches a **from the left** (that is, through values $x < a$), the values $f(x)$ come closer and closer to L; and

$$\lim_{x \to a^+} f(x) = R$$

means the same for x approaching a **from the right** ($x > a$). If both exist and

$$L = R,$$

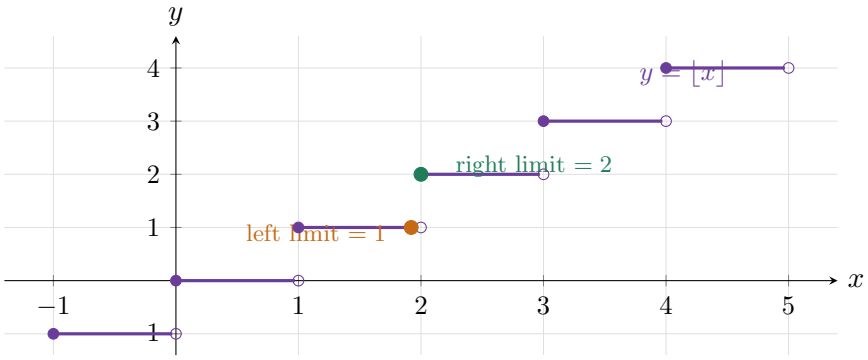
then $\lim_{x \to a} f(x)$ exists and equals that common value. Otherwise the limit **does not exist**.

Recall also that the limit need not depend on the value of f at a — indeed f need not even be defined at a.

Example 1.1 — the floor function again

For $g(x) = \lfloor x \rfloor$ and a an integer, say $a = 2$:

$$\lim_{x \to 2^-} \lfloor x \rfloor = 1, \qquad \lim_{x \to 2^+} \lfloor x \rfloor = 2, \qquad \lim_{x \to 2} \lfloor x \rfloor \text{ does not exist.}$$



At every integer the two one-sided limits disagree, so the limit fails to exist there; at every non-integer point they agree and the limit exists.

2 Continuity at a point

Definition 1.1 — Continuity at a point

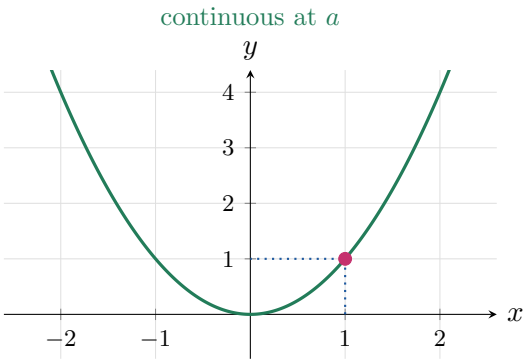
Let f be a function and let a be a point **in the domain of** f. Then f is **continuous at** a if

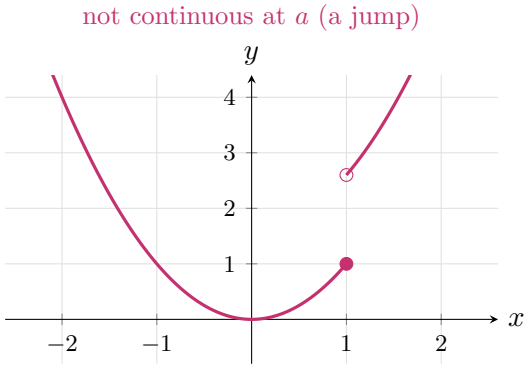
$$\boxed{\lim_{x \to a} f(x) = f(a)}$$

Three things are packed into this one line, and all three must hold:

1. $f(a)$ must be defined, i.e. a lies in the domain;
2. $\lim_{x \to a} f(x)$ must exist (so both one-sided limits exist and agree);
3. the two must be **equal**.

So continuity says that the limit at a may be found simply by **evaluating** the function at a — no jumps, no surprises.





3 Rules for limits of functions

Computing a limit directly from the definition — checking *all* sequences approaching the point — is impossible in practice. As with sequences, we instead build a stock of known limits together with rules for combining them.

Rules for limits

Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, and let $c \in \mathbb{R}$. Then

$$\lim_{x\to a}\big(f(x)+g(x)\big) = L+M \qquad \lim_{x\to a}\big(f(x)-g(x)\big) = L-M$$

$$\lim_{x\to a}\big(c\,f(x)\big) = c\,L \qquad \lim_{x\to a}\big(f(x)\,g(x)\big) = L\,M$$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{L}{M} \quad \textbf{provided } M \neq 0 \qquad \lim_{x\to a}\big(f(x)\big)^k = L^k \quad (k \in \mathbb{N})$$

The quotient rule needs care

In the quotient rule the denominator's limit M must be **non-zero**. If $M = 0$ the rule says nothing at all — and, as we saw, a $\frac{0}{0}$ situation may still have a perfectly good limit once the expression is rewritten.

The sandwich principle for functions

Suppose that near the point a we have

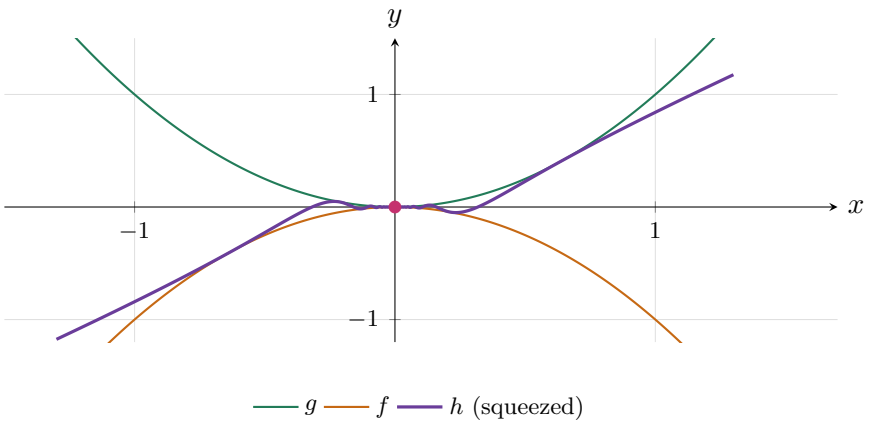
$$f(x) \;\le\; h(x) \;\le\; g(x),$$

and that

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = L.$$

Then h is squeezed between them and

$$\lim_{x\to a} h(x) = L.$$



3.1 A first consequence

Since $\lim_{x\to a} x = a$ and $\lim_{x\to a} c = c$, repeated use of the product and sum rules gives, for any polynomial p,

$$\lim_{x\to a} p(x) = p(a).$$

Polynomials are continuous

Every polynomial function is **continuous at every point** of $\mathbb{R}$.

The same argument then makes every **rational function** $\dfrac{p(x)}{q(x)}$ continuous at every point where $q(a) \neq 0$.

Example 1.2

Compute

$$\lim_{x\to 3} \frac{x^2 - 2x + 5}{x + 1}.$$

Both numerator and denominator are polynomials, so their limits are obtained by evaluation:

$$\lim_{x\to 3} \left(x^2 - 2x + 5\right) = 9 - 6 + 5 = 8, \qquad \lim_{x\to 3} (x + 1) = 4 \neq 0.$$

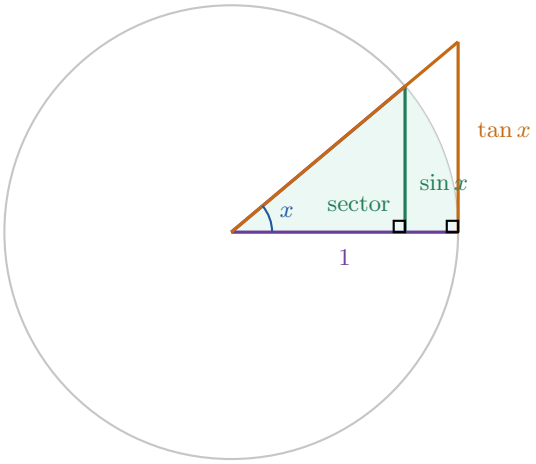
The denominator's limit is non-zero, so the quotient rule applies:

$$\lim_{x\to 3} \frac{x^2 - 2x + 5}{x + 1} = \frac{8}{4} = \boxed{2}$$

4 The standard limit $\lim_{x\to 0} \frac{\sin x}{x} = 1$

We met this limit before and verified it numerically. Here is the **geometric proof**, using areas and the sandwich principle.

Work in radians and take the **unit circle**. For $0 < x < \frac{\pi}{2}$, consider three regions: the inner triangle, the circular sector, and the outer right triangle.



Comparing the three areas:

$$\underbrace{\frac{1}{2}\sin x}_{\text{inner triangle}} \leq \underbrace{\frac{x}{2}}_{\text{sector of angle } x} \leq \underbrace{\frac{1}{2}\tan x}_{\text{outer triangle}}.$$

Where the sector area comes from

The area of a full unit circle is π, and a sector of angle x radians is the fraction $\frac{x}{2\pi}$ of it, giving

$$\pi \times \frac{x}{2\pi} = \frac{x}{2}.$$

The two triangles have area $\frac{1}{2} \times \text{base} \times \text{height}$, with base 1 and heights $\sin x$ and $\tan x$ respectively.

Multiply throughout by 2 and divide by $\sin x$ (which is positive for $0 < x < \frac{\pi}{2}$):

$$1 \;\le\; \frac{x}{\sin x} \;\le\; \frac{1}{\cos x},$$

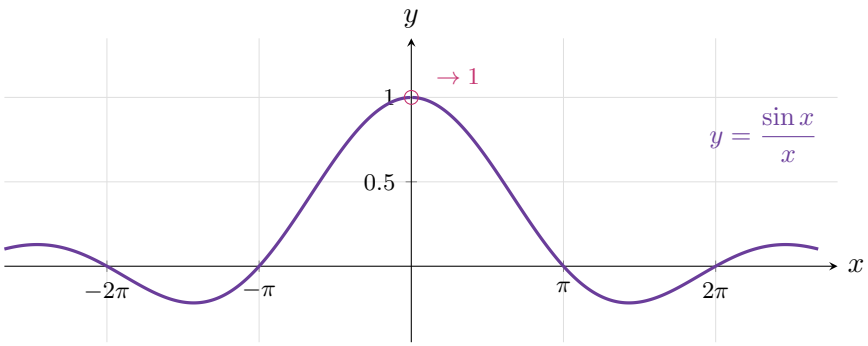
and taking reciprocals reverses the inequalities:

$$\cos x \;\le\; \frac{\sin x}{x} \;\le\; 1.$$

Now $\cos x \to 1$ as $x \to 0$, and the right-hand bound is the constant 1. By the **sandwich principle** the quantity in the middle is trapped, so

$$\boxed{\lim_{x\to 0} \frac{\sin x}{x} = 1}$$

The same bounds hold for $-\frac{\pi}{2} < x < 0$, since $\frac{\sin x}{x}$ is unchanged when x is replaced by $-x$.



5 A limit built from the previous one

Example 1.3

Compute

$$\lim_{x\to 0} \frac{1-\cos x}{x^2}.$$

Substituting $x = 0$ gives $\frac{1-1}{0} = \frac{0}{0}$, so we cannot substitute. Instead use a trigonometric identity. From

$$\cos 2\theta = 1 - 2\sin^2\theta \qquad \Longrightarrow \qquad 1 - \cos 2\theta = 2\sin^2\theta,$$

and putting $2\theta = x$, i.e. $\theta = \frac{x}{2}$,

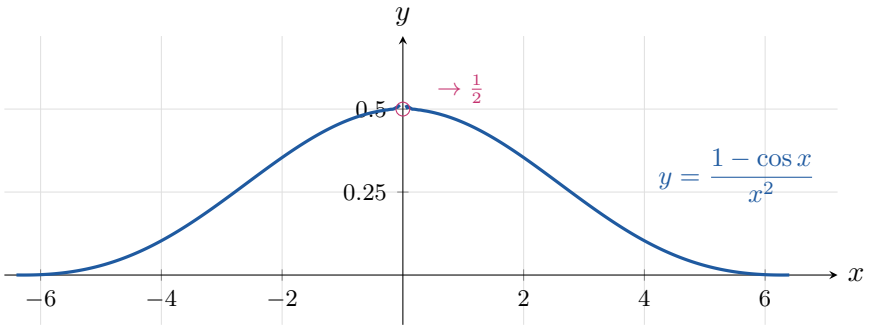
$$1 - \cos x = 2\sin^2\!\left(\frac{x}{2}\right).$$

Therefore

$$\frac{1-\cos x}{x^2} = \frac{2\sin^2\left(\frac{x}{2}\right)}{x^2} = \frac{1}{2}\cdot\frac{\sin^2\left(\frac{x}{2}\right)}{\left(\frac{x}{2}\right)^2} = \frac{1}{2}\left[\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right]^2.$$

As $x \to 0$ we also have $\frac{x}{2} \to 0$, so the bracket tends to 1 by the previous section, and by the product rule

$$\boxed{\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}}$$



6 The logarithmic limit

Example 1.4

Show that

$$\lim_{x\to 0}\frac{\log(1+x)}{x}=1.$$

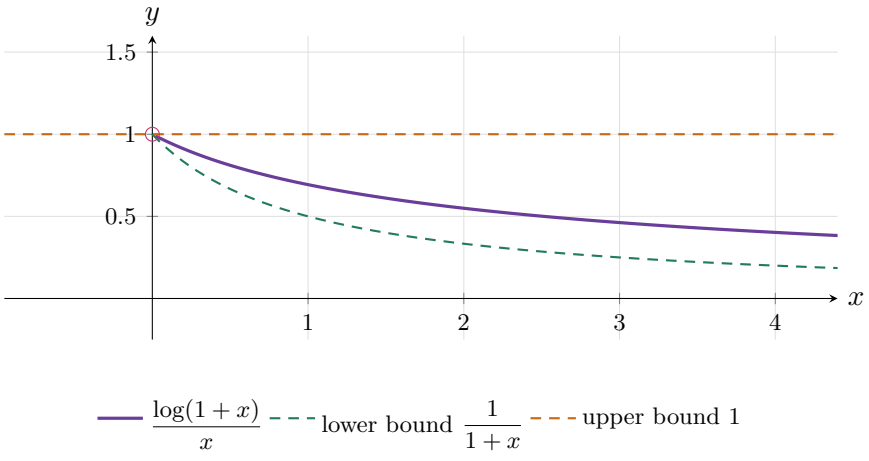
Again this is a $\frac{0}{0}$ situation. The proof rests on the inequalities, valid for $x>0$,

$$\frac{1}{1+x}\;\le\;\frac{\log(1+x)}{x}\;\le\;1.$$

(For $-1<x<0$ the two bounds simply exchange roles, becoming $1\le\frac{\log(1+x)}{x}\le\frac{1}{1+x}$, so the argument below covers that side too.)

As $x\to 0$ the left-hand bound $\dfrac{1}{1+x}$ tends to $\dfrac{1}{1}=1$, and the right-hand bound is the constant 1. Both outer quantities tend to 1, so by the **sandwich principle** the quantity in the middle is forced to follow:

$$\boxed{\lim_{x\to 0}\frac{\log(1+x)}{x}=1}$$



7 Limits and composition

A very useful consequence of continuity is that a limit may be moved **inside** a continuous function.

Pushing the limit inside

Suppose

$$\lim_{x\to a} g(x) = b,$$

and suppose f is **continuous at** b. Then the limit of the composition exists and

$$\lim_{x\to a} f\big(g(x)\big) = f(b) = f\Big(\lim_{x\to a} g(x)\Big).$$

So for a continuous f we may simply swap the symbols lim and f. In particular, a composition of continuous functions is continuous.

Example 1.5

$$\lim_{x\to 0} \sin\left(\frac{\log(1+x)}{x}\right) = \sin\left(\lim_{x\to 0} \frac{\log(1+x)}{x}\right) = \sin 1,$$

since sin is continuous everywhere and the inner limit is 1 by Example 1.4.

8 Which functions are continuous?

Standard continuous functions

Function	Continuous on
Polynomials	all of $\mathbb{R}$
Rational functions $\dfrac{p(x)}{q(x)}$	wherever $q(x) \neq 0$
$\sin x$, $\cos x$	all of $\mathbb{R}$
$\tan x$	wherever $\cos x \neq 0$
e^x, a^x $(a > 0)$	all of $\mathbb{R}$
$\log x$	$(0, \infty)$
x^k (power functions)	wherever defined

Since sums, products, quotients (with non-vanishing denominator) and compositions of continuous functions are again continuous, almost every function written down by a single formula is continuous wherever that formula makes sense. The interesting points are exactly the ones where it does *not*: the jumps, the zeros of a denominator, and the ends of a domain.

Continuity on an interval

When we say f is continuous on an interval, we mean it is continuous at each interior point. At an **endpoint** of the interval only one side is available, so there we ask only for the corresponding **one-sided** limit to equal the function value. We will not discuss continuity at isolated points, where no sequence can approach the point from within the domain at all.

Lecture 2 — Differentiability and the Derivative

Average versus instantaneous rate of change; the definition; worked examples; the derivative function

1 Average speed versus instantaneous speed

Let us begin by recalling the idea of a **rate of change** through an example.

Example 2.1 — the truck

A truck travels from Jalandhar in Punjab to Tiruchirappalli (Trichy) in Tamil Nadu. The distance is about 2900 km and the journey takes about 72 hours. What is its speed in km/h?

Most of us would divide the distance covered by the time taken:

$$\text{speed} = \frac{2900 \text{ km}}{72 \text{ h}} \approx 40 \text{ km/h}.$$

The fine near Nagpur

After reaching Trichy the driver tells his friend that he was fined by a traffic constable near Nagpur, Maharashtra — the route passes Agra, Jhansi, Nagpur, Hyderabad and so on — for **speeding**.

The friend calculates the speed from the data above, gets 40 km/h, and declares that the constable was wrong and the fine unjustified, since the highway speed limit is above 40 km/h.

Is the friend correct?

From this information, can we conclude that the truck's speed *at the moment the constable stopped it* was 40 km/h? Surely not.

New information

The driver then mentions that he covered the 260 km stretch within Maharashtra — entering from Madhya Pradesh and leaving into Telangana — in 4 hours, of which about 1 hour was a breakfast break. So the truck was actually moving for 3 hours:

$$\text{speed within Maharashtra} = \frac{260}{3} \approx 86.67 \text{ km/h},$$

well above the speed limit. The friend recalculates, withdraws his opinion, and now agrees the fine was justified.

Is the friend correct *now*?

No — and for exactly the same reason as before. If the first argument was wrong, so is the second: we still cannot say what speed the truck was doing *at the instant* of the fine, because we are again reasoning from an **average**.

Stretch	Distance / time	Average speed
Jalandhar → Trichy (whole journey)	2900 km / 72 h	≈ 40 km/h
Within Maharashtra (excluding the 1 h break)	260 km / 3 h	≈ 86.67 km/h

The lesson

Average speed and instantaneous speed are different concepts, and one *cannot* deduce the instantaneous speed from the average. The speed increases and decreases along the way — which is precisely why a vehicle has a **speedometer**. What the speedometer shows cannot be computed by saying "I travelled this much distance in this much time".

From the given data there is simply no way to recover the instantaneous speed. A dashboard camera recording the speedometer needle would give it; so would the constable's **speed gun**, which reports the speed instantaneously.

2 Towards instantaneous speed

To calculate the instantaneous speed at some time, we take the distance travelled in a very **short** period of time around that instant and divide by that period. For example, if the truck travels y km in the next 1 minute, then the instantaneous speed is approximately

$$60\,y \text{ km/h}.$$

Why only *approximately*? Because even one minute is not instantaneous: within a minute there may be a series of speed breakers, a hill, an accident, or a stop for chai. Still, we feel intuitively that this is a better approximation than one based on the whole journey.

Ideally we take the time interval as small as we possibly can — an **infinitesimal** time. And shrinking an interval to zero is exactly what a **limit** does.

Instantaneous speed as a limit

To find the speed at time t_0, take the distance travelled between t_0 and $t_0 + \Delta t$, divide by Δt, and let $\Delta t \to 0$:

$$\text{instantaneous speed at } t_0 = \lim_{\Delta t \to 0} \frac{f(t_0 + \Delta t) - f(t_0)}{\Delta t},$$

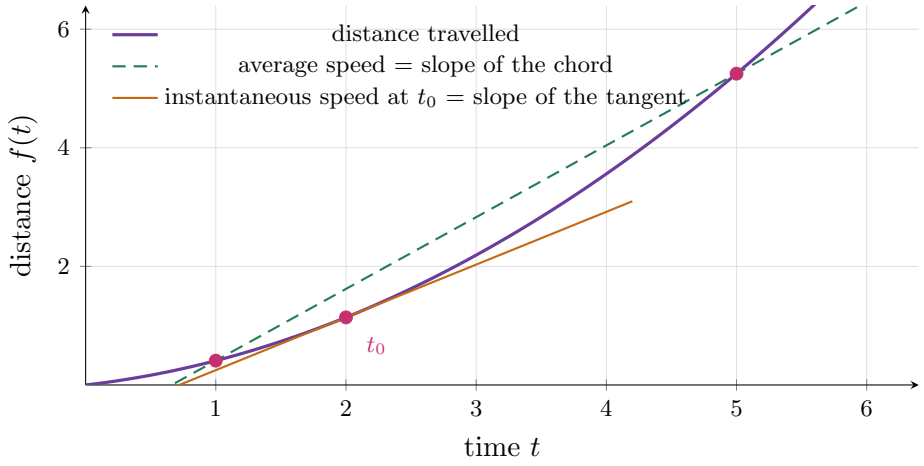
where f is the distance travelled as a function of time.

Mind the units

The measurements must be consistent. For km/h, Δt must be in hours and the distance in kilometres. So if Δt is 1 second and the distance is 1 metre, think of them as

$$1 \text{ m} = 10^{-3} \text{ km}, \qquad 1 \text{ s} = \frac{1}{3600} \text{ h}.$$

This is one of the main reasons we were interested in limits in the first place: limits sit at the heart of anything *instantaneous*.



The chord through two points gives an **average** rate of change; letting the second point slide into the first turns the chord into the **tangent**, whose slope is the instantaneous rate.

3 Differentiability: the definition

Speed is the rate of change of distance with respect to time. Abstracting from the example, the concept we need is **differentiability**.

Definition 2.1 — Differentiability at a point

Let f be a function defined on an open interval around a point a. Then f is **differentiable at** a if

$$\boxed{\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} \text{ exists.}}$$

When it exists, this limit is the **instantaneous rate of change** of f at a.

Comparing with the truck: there f was the distance covered and the variable was time, so writing the instantaneous speed at t_0 we had

$$\lim_{\Delta t\to 0}\frac{f(t_0+\Delta t)-f(t_0)}{\Delta t}.$$

Here Δt has been renamed h and t_0 renamed a.

Think of it as a function of h

When we take $\lim_{h\to 0}$, the whole expression $\dfrac{f(a+h)-f(a)}{h}$ must be regarded as a **function of the variable** h. That is the function whose limit we are computing.

4 Examples: differentiable

Example 2.2

Show that $f(x)=x$ is differentiable at every point a.

Here $f(a+h)=a+h$ and $f(a)=a$, so

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}=\lim_{h\to 0}\frac{(a+h)-a}{h}=\lim_{h\to 0}\frac{h}{h}=\lim_{h\to 0}1=\boxed{1}$$

So the rate of change of $f(x) = x$ is 1 — not a difficult fact to appreciate: move 1 unit along and the function changes by 1.

Example 2.3

Show that $f(x) = \sin x$ is differentiable at 0.

$$\lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{\sin(0+h) - \sin 0}{h} = \lim_{h\to 0} \frac{\sin h - 0}{h} = \lim_{h\to 0} \frac{\sin h}{h} = \boxed{1}$$

using the limit for which we gave a detailed argument in the previous lecture. So $\sin x$ is differentiable at 0, and the value of the limit is 1.

5 Examples: not differentiable

5.1 The modulus function at 0

Example 2.4

Show that $f(x) = |x|$ is not differentiable at 0.

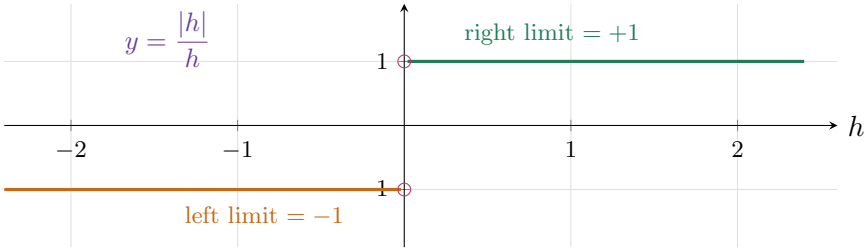
This is one of the places where we got stuck when discussing tangents. Compute:

$$\lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{|h| - |0|}{h} = \lim_{h\to 0} \frac{|h|}{h}.$$

Now look at the function $\dfrac{|h|}{h}$:

$$\frac{|h|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0, \end{cases}$$

and it is not defined at $h = 0$ — but that does not matter, since we only want the limit.



From the left the value is -1; from the right it is $+1$. The two one-sided limits disagree, so the limit **does not exist** — and therefore $|x|$ is **not differentiable at** 0.

5.2 The cube root at 0

Example 2.5

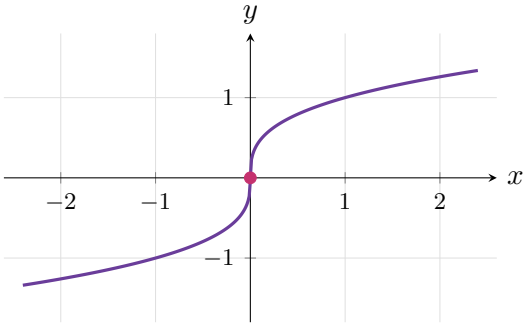
Show that $f(x) = x^{1/3}$ is not differentiable at 0.

From first principles,

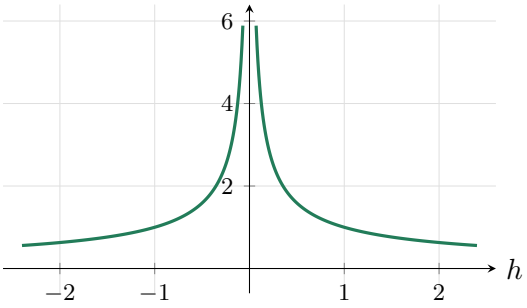
$$\lim_{h\to 0} \frac{h^{1/3} - 0^{1/3}}{h} = \lim_{h\to 0} \frac{h^{1/3}}{h} = \lim_{h\to 0} h^{-2/3} = \lim_{h\to 0} \frac{1}{\sqrt[3]{h^2}}.$$

Written that way it is clear: square h, take the cube root, and divide by it. As h becomes small the denominator becomes tiny, so the quotient becomes very large. The limit therefore **does not exist** — it diverges to infinity, from the right *and* from the left, since h^2 is positive either way. This is the same behaviour we met for $\frac{1}{h}$.

$y = x^{1/3}$: a vertical tangent at 0



the quotient $h^{-2/3} \to \infty$



5.3 The floor function at an integer

Example 2.6

Show that $f(x) = \lfloor x \rfloor$ is not differentiable at any integer point.

Take the integer 0 (the same argument works at every integer):

$$\lim_{h\to 0} \frac{\lfloor 0+h \rfloor - \lfloor 0 \rfloor}{h} = \lim_{h\to 0} \frac{\lfloor h \rfloor}{h}, \qquad \text{since } \lfloor 0 \rfloor = 0.$$

Now

$$\frac{\lfloor h \rfloor}{h} = \begin{cases} 0, & 0 < h < 1, \\ \dfrac{-1}{h}, & -1 < h < 0. \end{cases}$$

From the right the limit is 0. From the left we have $\dfrac{-1}{h}$ with h negative and small, which **diverges to $+\infty$** — note the sign: $\frac{1}{h}$ would go to $-\infty$, but the extra minus flips it. Either way the left-hand limit does not exist, so the limit does not exist and $\lfloor x \rfloor$ is not differentiable at the integers.

Where the earlier results paid off

Every one of these three verdicts came from limit results proved in the previous lectures — the one-sided limits of a step function, and the divergence of $\frac{1}{h}$ at 0. Those results are now doing the work of checking differentiability for us.

6 Differentiability implies continuity

Theorem 2.1

If f is differentiable at a, then f is continuous at a.

Proof

We want to conclude that $\lim_{x\to a} f(x) = f(a)$.

We know, by hypothesis, that

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}=L$$

exists. We also know the elementary limit

$$\lim_{h\to 0} h = 0.$$

Multiply the two functions. By the product rule for limits — if two functions have limits at a point, the limit of the product is the product of the limits — we get

$$\lim_{h\to 0}\left[\frac{f(a+h)-f(a)}{h}\cdot h\right] = L\times 0 = 0,$$

that is,

$$\lim_{h\to 0}\big(f(a+h)-f(a)\big)=0.$$

Now $f(a)$ is a constant, so adding it to both sides gives

$$\lim_{h\to 0} f(a+h) = f(a).$$

Finally, writing $x = a+h$ — so that $h\to 0$ means exactly $x\to a$ — this reads

$$\lim_{x\to a} f(x) = f(a),$$

which is precisely continuity at a. ■

Differentiability is the stronger notion

Differentiability is a much more powerful concept than continuity: if a function is differentiable at a point, it is certainly continuous there. The converse fails — $|x|$ is continuous at 0 but not differentiable there.

7 The derivative function

Definition 2.2 — The derivative

For a function f, the **derivative function**, denoted $f'(x)$ or $\dfrac{df}{dx}(x)$, is

$$\boxed{f'(x)=\frac{df}{dx}(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}}$$

— the value of that limit at the point x.

Evaluating at different points gives different values, and that assignment is exactly the derivative function. At some points the derivative may fail to exist: for the floor function it does not exist at the integers, but it exists everywhere else, so we may speak of the derivative function of $\lfloor x\rfloor$ at all non-integer points.

The **domain of the derivative function** is the set of points at which f is differentiable.

7.1 Example: $f(x) = x$

We already computed the limit at an arbitrary point and got 1, so

$$f(x) = x \qquad \Longrightarrow \qquad f'(x) = 1.$$

7.2 Example: $f(x) = \sin x$

We know $f'(0) = 1$; let us now do an arbitrary point a:

$$f'(a) = \lim_{h \to 0} \frac{\sin(a+h) - \sin a}{h}.$$

Use the expansion $\sin(a+h) = \sin a \cos h + \cos a \sin h$:

$$f'(a) = \lim_{h \to 0} \frac{\sin a \cos h + \cos a \sin h - \sin a}{h}.$$

Break it into two parts:

$$f'(a) = \lim_{h \to 0} \cos a \cdot \frac{\sin h}{h} \; + \; \lim_{h \to 0} \sin a \cdot \frac{\cos h - 1}{h}.$$

First limit. Here $\cos a$ is a constant, since the limit is over h, and $\dfrac{\sin h}{h} \to 1$. So the first term is $\cos a$.

Second limit. Recall that $\lim\limits_{h \to 0} \dfrac{1 - \cos h}{h^2} = \dfrac{1}{2}$. Pulling $\sin a$ out and multiplying and dividing by h,

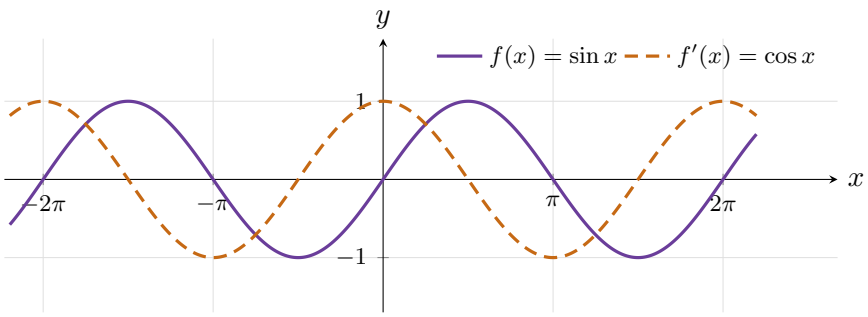
$$\frac{\cos h - 1}{h} = -\,h \cdot \frac{1 - \cos h}{h^2} \xrightarrow[h \to 0]{} -\,0 \times \frac{1}{2} = 0.$$

So the second term is $\sin a \times 0 = 0$.

Adding,

$$\boxed{f(x) = \sin x \qquad \Longrightarrow \qquad f'(x) = \cos x}$$

and we have demonstrated this with a proof.



Everything in this lecture was computed from **first principles** — writing the limit and evaluating it. As we did for continuity, we shall next establish **rules for derivatives** and then use those rules to compute further derivatives without returning to the definition each time.

8 Back to tangents

Why did we do all this? We were studying **tangents**, and two examples gave us pause: the floor function and $y = |x|$.

- Demanding **continuity** before speaking of tangents disposes of the **floor function** — it is not continuous at the integers.
- But continuity was **not good enough** to dispose of $y = |x|$: that function is perfectly continuous at 0, and yet has no tangent there.
- Demanding **differentiability** disposes of it too, since $|x|$ is not differentiable at 0.

> We shall speak of **tangents only at points where the function is differentiable**.

Note that this does not throw the whole function away: for $|x|$ at any point other than 0 we are perfectly happy to talk about the tangent — and, as the earlier lecture on tangents showed, there was never a problem away from 0. The problem was exactly at 0.

Lecture 3 — Computing Derivatives and L'Hôpital's Rule

Linearity, product, quotient and chain rules; worked examples; indeterminate limits

1 Recalling the derivative

Differentiability and the derivative

Let f be a function defined on an open interval around the point a. Then f is **differentiable at** a if

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} \quad \text{exists,}$$

and at every point x where this limit exists we call its value $f'(x)$ or $\dfrac{df}{dx}(x)$. This defines a new function — the **derivative function**, or simply the derivative of f.

The limit may fail to exist at some points, so the derivative function has domain consisting of exactly those points at which f is differentiable.

A convention

Colloquially one says "f *is differentiable*" — without qualifying at which points — to mean that f is differentiable **everywhere**. In that case the derivative function is defined at every point where f itself is defined.

Recall also the fact proved last time:

$$f \text{ differentiable at } a \quad \Longrightarrow \quad f \text{ continuous at } a.$$

It was this fact that let us dispose of the two problematic cases we met when discussing tangents: $|x|$, which has a corner, and $\lfloor x \rfloor$, which is discontinuous.

2 Rules for derivatives

Since the derivative is defined in terms of limits, it is natural that the rules for limits should lead to rules for derivatives.

2.1 Linearity

Linearity

If f and g are differentiable at a, and $c \in \mathbb{R}$, then $f+g$, cf and $f-g$ are all differentiable at a, and

$$(f+g)'(a) = f'(a)+g'(a), \qquad (cf)'(a) = c\,f'(a), \qquad (f-g)'(a) = f'(a)-g'(a).$$

The first two follow directly from the corresponding rules for limits; putting them together with $c=-1$ gives the third.

2.2 The product rule

Here things start to differ from what we have seen before. For limits and continuity, limits simply got multiplied — but here we are not taking the limit of the functions themselves; we are taking the limit of $\dfrac{f(x+h)-f(x)}{h}$, which is a much trickier object.

Product rule

If f and g are differentiable at a, then so is fg, and

$$\boxed{(fg)'(a) = f'(a)\,g(a) + f(a)\,g'(a)}$$

Why it works — the idea

The quantity to compute is

$$\lim_{h\to 0} \frac{f(a+h)\,g(a+h) - f(a)\,g(a)}{h}.$$

We know each function is differentiable individually, so we **add and subtract** a suitable term in the middle:

$$\frac{f(a+h)\big[g(a+h)-g(a)\big] \;+\; \big[f(a+h)-f(a)\big]\,g(a)}{h}.$$

Splitting this into two separate limits and passing to the limit gives exactly the identity above.

Remember the identity

Whether or not you follow every step of the proof, **do remember the identity** — we shall use it repeatedly in the examples.

2.3 The quotient rule

Quotient rule

If f and g are differentiable at a and $g(a) \neq 0$, then

$$\boxed{\left(\frac{f}{g}\right)'(a) = \frac{f'(a)\,g(a) - f(a)\,g'(a)}{\big[g(a)\big]^2}}$$

The idea is similar to the product rule, but the expression must be juggled a little more. Those of a more mathematical inclination are encouraged to write out the limit and rearrange it — it is not particularly difficult, but it does need some finesse. Note that $g(a)$ sits in the denominator, so it had better be non-zero.

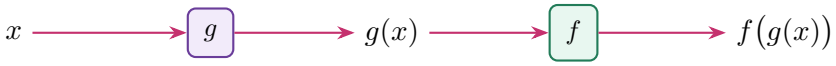
2.4 The chain rule (composition rule)

Chain rule

If f and g are differentiable, then

$$\boxed{(f\circ g)'(x) = f'\big(g(x)\big)\cdot g'(x)}$$

In words: start differentiating from the **outside**, keeping whatever is inside intact; then move inside, take its derivative, and multiply.



$$\text{derivative} = \underbrace{f'(g(x))}_{\text{outside first}} \times \underbrace{g'(x)}_{\text{then inside}}$$

(The name *chain rule* will be explained later in the course, when the same rule reappears in higher dimensions.)

3 Baby examples: powers of x

We already know from first principles that $f(x) = x$ has $f'(x) = 1$. Let us build up from there using the **product rule**.

Example 3.1

Find the derivatives of x^2, x^3, and then of x^n.

For x^2: write $x^2 = x \cdot x$, so

$$\frac{d}{dx}\left(x^2\right) = \underbrace{1}_{(x)'} \cdot x + x \cdot \underbrace{1}_{(x)'} = x + x = 2x.$$

For x^3: write $x^3 = x^2 \cdot x$, so

$$\frac{d}{dx}\left(x^3\right) = \underbrace{2x}_{(x^2)'} \cdot x + x^2 \cdot 1 = 2x^2 + x^2 = 3x^2.$$

The pattern is now clear, and the guess is:

$$f(x) = x^n \qquad \Longrightarrow \qquad f'(x) = n\,x^{n-1}.$$

Proof by induction

Assume the result for $n-1$. Write $x^n = x^{n-1} \cdot x$ and apply the product rule:

$$\frac{d}{dx}\left(x^n\right) = \underbrace{(n-1)\,x^{n-2}}_{\text{by induction}} \cdot x \;+\; x^{n-1} \cdot 1 = (n-1)\,x^{n-1} + x^{n-1} = n\,x^{n-1}. \blacksquare$$

Here n is a natural number.

The general power rule

$$f(x) = x^a \qquad \Longrightarrow \qquad f'(x) = a\,x^{a-1}$$

for a general exponent a. This one has to be done from first principles, and comes with caveats: once a is allowed to be negative one must be careful about what happens at $x = 0$.

4 Polynomials

Example 3.2

Differentiate

$$f(x) = 5x^3 - 17x^2 + \pi x - 0.5.$$

By **linearity** we may handle each monomial separately, pulling the constants out:

$$\begin{aligned} f'(x) &= 5 \cdot \frac{d}{dx}\left(x^3\right) - 17 \cdot \frac{d}{dx}\left(x^2\right) + \pi \cdot \frac{d}{dx}(x) - \frac{d}{dx}(0.5) \\ &= 5\left(3x^2\right) - 17\,(2x) + \pi(1) - 0 \\ &= \boxed{15x^2 - 34x + \pi} \end{aligned}$$

The derivative of a constant is 0

This is immediate from first principles. (One may also think of the constant as a multiple of x^0 and apply the power rule to get $0 \times x^{-1}$ — not really the correct way to think about it, but either route gives 0.)

General polynomial

If

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

then

$$f'(x) = n\,a_n x^{n-1} + (n-1)\,a_{n-1}x^{n-2} + \cdots + a_1.$$

Note that the constant term a_0 **drops out**, and the last surviving term is a_1, since $a_1 x$ differentiates to a_1.

5 A product of two familiar functions

Example 3.3

Differentiate $f(x) = x^7 \sin x$.

Apply the product rule, using $\frac{d}{dx}\left(x^7\right) = 7x^6$ and $\frac{d}{dx}(\sin x) = \cos x$ (proved from first principles last lecture):

$$f'(x) = 7x^6 \sin x + x^7 \cos x.$$

So even a function that looks complicated has a derivative that is not hard to compute: if you know the derivatives of the individual pieces appearing in the expression, you can usually put them together to get the derivative of the whole.

6 The quotient rule at work

Example 3.4

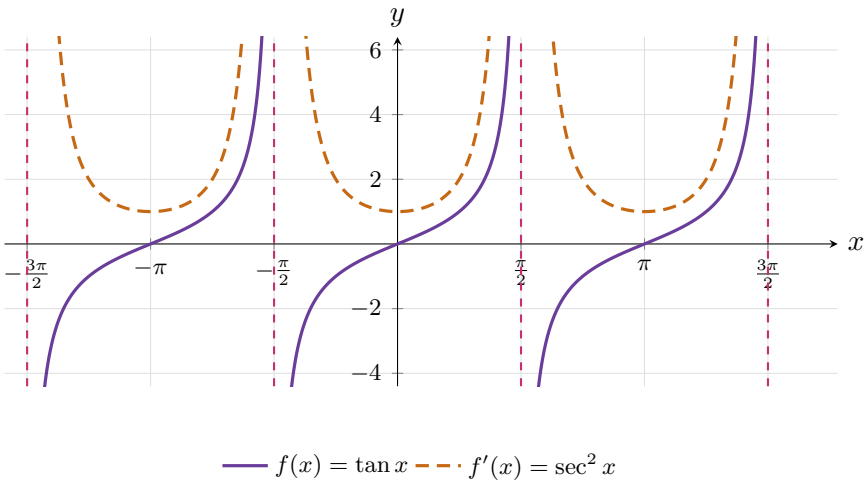
Differentiate $f(x) = \tan x$.

Warning bells already: $\tan x$ is not defined at $\pm\frac{\pi}{2}$, so everything below makes sense only where the function is well behaved.

Write $\tan x = \dfrac{\sin x}{\cos x}$ and apply the quotient rule with $f = \sin$, $g = \cos$, remembering that $\frac{d}{dx}(\cos x) = -\sin x$:

$$\begin{aligned}\frac{d}{dx}(\tan x) &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \boxed{\sec^2 x}\end{aligned}$$

using the familiar identity $\sin^2 x + \cos^2 x = 1$ and the definition $\sec x = \frac{1}{\cos x}$.
This is valid at all x except the odd multiples of $\frac{\pi}{2}$, where $\cos x = 0$ and the expression makes no sense.



Example 3.5

Differentiate $f(x) = \dfrac{1}{x^r}$ (for $x \neq 0$).

Take the numerator to be the constant 1 and the denominator x^r, assuming the power rule for positive r. Since $\frac{d}{dx}(1) = 0$, the first term contributes nothing:

$$\frac{d}{dx}\left(\frac{1}{x^r}\right) = \frac{0 \cdot x^r - 1 \cdot r\,x^{r-1}}{(x^r)^2} = \frac{-r\,x^{r-1}}{x^{2r}} = -r\,x^{r-1-2r} = \boxed{-r\,x^{-r-1}}$$

Rewriting $\dfrac{1}{x^r} = x^{-r}$, this says exactly that

$$\frac{d}{dx}\left(x^{-r}\right) = (-r)\,x^{-r-1},$$

so the negative powers obey the **same** rule $\frac{d}{dx}(x^n) = nx^{n-1}$ that we proved for natural numbers.

7 The chain rule at work

Example 3.6

Differentiate $f(x) = \tan(2x)$.

There are several routes: one could write it as $\dfrac{\sin 2x}{\cos 2x}$, or use the identity $\tan 2x = \dfrac{2\tan x}{1-\tan^2 x}$. Let us instead use the **composition rule**.

Set

$$g(x) = 2x, \qquad h(x) = \tan x, \qquad \text{so that} \qquad f(x) = h\big(g(x)\big).$$

Then

$$f'(x) = h'\big(g(x)\big) \cdot g'(x).$$

We just computed $h'(x) = \sec^2 x$, so $h'\big(g(x)\big) = \sec^2(2x)$; and $g'(x) = 2$. Hence

$$\boxed{\frac{d}{dx}\big(\tan 2x\big) = 2\sec^2(2x)}$$

You are encouraged to check that either of the other two routes gives the same answer.

8 Indeterminate limits

Now for an application of derivatives — to computing **limits**, which is a pleasing interplay, since limits were what we used to define derivatives in the first place.

Indeterminate form

Let f and g be defined on an interval around a, and suppose that

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0,$$

or both limits diverge to $+\infty$, or both diverge to $-\infty$. Then the quotient $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)}$ is said to be of **indeterminate form**:

$$\frac{0}{0} \qquad \text{or} \qquad \frac{\infty}{\infty}.$$

Recall the situation. If both limits exist and the denominator's limit is *non-zero*, the quotient rule for limits applies. If the denominator's limit is 0 but the numerator's is not, the limit does not exist. But if **both** are 0 — or both infinite — we do not know what happens, and the quotient rule cannot be used at all.

9 L'Hôpital's Rule

About the name

It is pronounced *lo-pee-TAL*: the H is silent, and the word after the L is the French for *hospital*. L'Hôpital was a French mathematician — though the rule is quite unfortunately named after him, since it was very likely discovered by his mentor, **Bernoulli**, whom he possibly paid to have it named after himself. (You may recognise the name Bernoulli from statistics.)

L'Hôpital's Rule

In the situation of an indeterminate form, suppose that

1. f' and g' exist on the interval, except possibly at a;
2. $g'(x) \neq 0$ on the interval, except possibly at a;
3. $\lim\limits_{x\to a} \dfrac{f'(x)}{g'(x)} = L$ — it is essential that the limit of the quotient of the *derivatives* exists.

Then

$$\boxed{\lim_{x\to a}\frac{f(x)}{g(x)} = L}$$

So although we cannot apply the quotient rule to $\frac{f}{g}$, we may instead differentiate the numerator and the denominator *separately* and take the limit of that quotient — and the answer is the same L. This seems like something of a miracle.

9.1 Example: the logarithmic limit

Example 3.7

$$\lim_{x\to 0}\frac{\log(1+x)}{x}$$

First check the form. Since log is continuous we may substitute:

$$\lim_{x\to 0}\log(1+x) = \log(1+0) = \log 1 = 0,$$

and the denominator's limit is clearly 0. So this is a $\frac{0}{0}$ situation and the quotient rule is unavailable. Apply L'Hôpital. The derivative of the numerator is $\dfrac{1}{1+x}$ (obtained from the derivative of $\log x$ together with the composition rule), and the derivative of the denominator is 1:

$$\lim_{x\to 0}\frac{\log(1+x)}{x} = \lim_{x\to 0}\frac{\frac{1}{1+x}}{1} = \lim_{x\to 0}\frac{1}{1+x} = \frac{1}{1+0} = \boxed{1}$$

— and here we really may substitute, since $\frac{1}{1+x}$ is continuous at 0.

9.2 More examples

Example 3.8

$$\lim_{x\to 2}\frac{x^2-5x+6}{x-2}$$

This is the $\frac{0}{0}$ form we met before. Differentiating numerator and denominator:

$$\lim_{x\to 2}\frac{2x-5}{1} = 2(2)-5 = 4-5 = \boxed{-1}$$

which is exactly what we obtained earlier by factorising into $x-3$.

Example 3.9

$$\lim_{x\to 0}\frac{\sin x}{x}$$

Earlier this needed a fairly involved geometric proof. With L'Hôpital, differentiate top and bottom:

$$\lim_{x\to 0}\frac{\cos x}{1} = \cos 0 = \boxed{1}$$

since cos is continuous and we may substitute.

Example 3.10

$$\lim_{x\to\infty} \frac{a + b\,e^x}{c + d\,e^x}$$

Limits at infinity

The rule was stated on an interval around a point a. We may allow that "point" to be infinity — though infinity is not a point of the real line, we think of it as sitting at the end — provided the same conditions hold for all x larger than some N. L'Hôpital then applies here too.

Here both numerator and denominator blow up, so this is the $\frac{\infty}{\infty}$ form. Differentiating:

$$\lim_{x\to\infty} \frac{b\,e^x}{d\,e^x} = \lim_{x\to\infty} \frac{b}{d} = \boxed{\frac{b}{d}}$$

after cancelling e^x; the result is a constant, so the limit is that constant. Compare this with our earlier route, where we multiplied everything through by e^{-x}.

Example 3.11

$$\lim_{x\to 0} \frac{1 - \cos x}{x^2}$$

Earlier we did this by trigonometric jugglery, expressing $\cos x$ in terms of $\sin\frac{x}{2}$. Instead, differentiate. Since $\frac{d}{dx}(\cos x) = -\sin x$, the derivative of the numerator is $-(-\sin x) = \sin x$; the derivative of the denominator is $2x$. So

$$\lim_{x\to 0} \frac{\sin x}{2x} = \frac{1}{2}\lim_{x\to 0} \frac{\sin x}{x} = \frac{1}{2} \times 1 = \boxed{\frac{1}{2}}$$

which is exactly what we obtained before.

Limit	Form	After differentiating	Value
$\frac{\log(1+x)}{x}$, $x \to 0$	$\frac{0}{0}$	$\frac{1/(1+x)}{1}$	1
$\frac{x^2 - 5x + 6}{x - 2}$, $x \to 2$	$\frac{0}{0}$	$\frac{2x - 5}{1}$	-1
$\frac{\sin x}{x}$, $x \to 0$	$\frac{0}{0}$	$\frac{\cos x}{1}$	1
$\frac{a + be^x}{c + de^x}$, $x \to \infty$	$\frac{\infty}{\infty}$	$\frac{be^x}{de^x}$	$\frac{b}{d}$
$\frac{1 - \cos x}{x^2}$, $x \to 0$	$\frac{0}{0}$	$\frac{\sin x}{2x}$	$\frac{1}{2}$

Do check the conditions

In the examples above the calculations were done quickly, going straight to the differentiation. Ideally one should first compute both derivatives and **check the three conditions** of the theorem, and only then apply it. Here those conditions do hold — you are encouraged to verify them.

So L'Hôpital's Rule is a very nice rule, and many indeterminate cases can be resolved with it quickly.

Lecture 4 — Derivatives, Tangents and Linear Approximation

The tangent as a limit of secants; the equation of the tangent line; best linear approximation

1 Where we are

In the earlier lectures we met the notion of a **tangent**, and found it hard to say when a tangent exists, when it does not, and how to compute it. That is precisely why we developed the theory: limits, then continuity, then differentiability, and then rules for computing derivatives. In this lecture we finally make the connection between **derivatives and tangents**, and along the way obtain a bonus — the idea of a **linear approximation**.

Recalling differentiability

Let f be defined on an open interval around a. Then f is **differentiable at** a if

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

exists, and the value of that limit is the **derivative** of f at a. Doing this at each x gives the derivative function, denoted f' or $\dfrac{df}{dx}$.

Two notations, two authors

Each notation has its advantages. It is a classic fact that one of them is due to **Newton** and the other to **Leibniz** — and that one of the two made life considerably easier than the other.

Recalling the tangent

For a function f and a point a in its domain, the **tangent to** f **at** a is the line representing the **instantaneous direction** in which the graph Γ_f moves at the point $\big(a, f(a)\big)$. Traditionally it is also described as the line that "just touches" the graph there.

Why a connection is not surprising

When we introduced differentiability, the motivating example was **instantaneous speed**: if f denotes the trajectory of a particle, the limit $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ is the instantaneous speed. The tangent is about instantaneous *direction*, the derivative about instantaneous *speed* — so it is hardly surprising that the two are connected.

2 The tangent as a limit of secants

Secant

A **secant** is a line obtained by joining two points of a curve.

The key idea

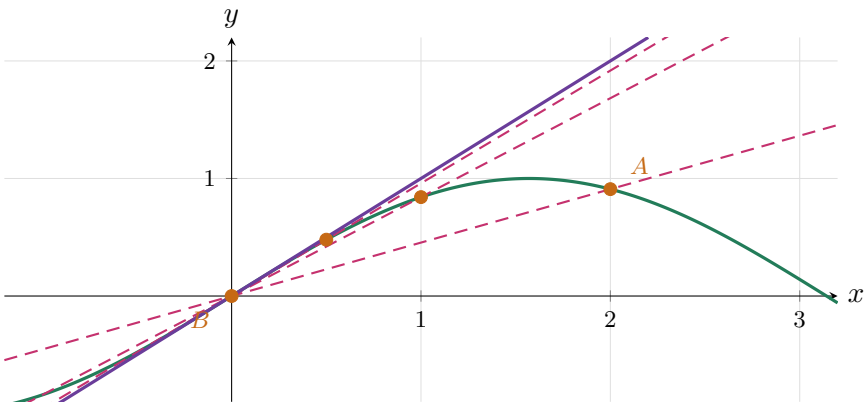
If the tangent to f at a exists — and this is the key proviso — then we may think of it as the **limit of secants** joining

$$\big(a,\, f(a)\big) \qquad \text{and the nearby point} \qquad \big(a+h,\, f(a+h)\big),$$

as h is allowed to come closer and closer to 0.

2.1 Watching it happen: $f(x) = \sin x$ at $a = 0$

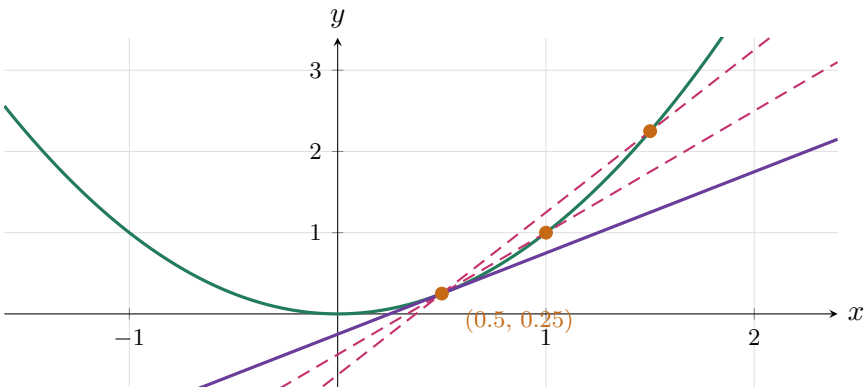
Take $B = (0, 0)$, since $\sin 0 = 0$, and start the second point at $A = (2, \sin 2) = (2,\, 0.91)$. As h shrinks, A slides along the graph towards B, and the secant AB moves towards the tangent — which here is the line $y = x$.



$f(x) = \sin x$ — — · secants, $h = 2,\ 1,\ 0.5$ — tangent $y = x$

2.2 $f(x) = x^2$ at $a = 0.5$

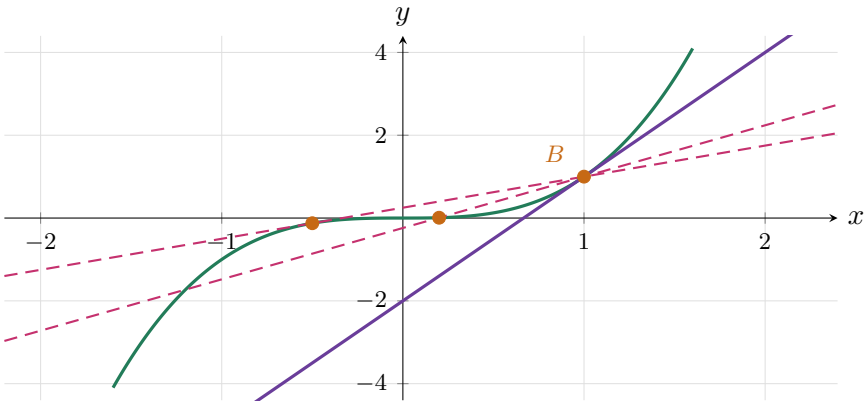
Here the point on the graph is $\big(0.5,\, 0.25\big)$, since the function is x^2. Joining it to neighbouring points gives secants which visibly close in on the tangent as the second point approaches.



$f(x) = x^2$ — — · secants $(h = 1,\ 0.5)$ — tangent $y = x - 0.25$

2.3 $f(x) = x^3$: the approach can be untidy

For x^3 something slightly strange can happen. As the second point moves in, the secant may first **cross over** the tangent and move away, before turning around and coming back to it.



$f(x) = x^3$ — secants — tangent at $x = 1$: $y = 3x - 2$

Why that is allowed

If the second point starts far away, the secant may first move away and only then come back. That is perfectly fine when we are taking **limits**: the requirement is only that the secant *eventually* comes closer and closer, not that it approaches monotonically.

3 The equation of the secant, and its limit

The line through the two points $\big(a, f(a)\big)$ and $\big(a+h, f(a+h)\big)$ has equation

$$y - f(a) = \underbrace{\frac{f(a+h) - f(a)}{(a+h) - a}}_{\text{slope}} (x - a) = \frac{f(a+h) - f(a)}{h}\,(x - a),$$

since the denominator simplifies to h. That slope should ring a bell.

Cross-checking that it passes through both points

Put $x = a$ and $y = f(a)$: both sides are 0, so the line passes through $\big(a, f(a)\big)$.
Put $x = a+h$ and $y = f(a+h)$: the right-hand side becomes $\dfrac{f(a+h) - f(a)}{h} \cdot h = f(a+h) - f(a)$, which is exactly the left-hand side. So the line passes through $\big(a+h, f(a+h)\big)$ as well.

Rewriting the secant equation as

$$y = \boxed{\frac{f(a+h) - f(a)}{h}}\,(x - a) + f(a),$$

we now let $h \to 0$. *Precisely when* the boxed quantity has a limit — that is, precisely when f is differentiable at a — the boxed quantity becomes $f'(a)$, and everything else is unchanged.

Theorem 4.1 — Tangents and derivatives

Let f be differentiable at a. Then the **tangent to f at a exists** and is given by

$$\boxed{y = f'(a)\,(x-a) + f(a)}$$

Conversely, if the tangent to f at a exists and is **not vertical** (i.e. it is not the line $x = a$), then f is differentiable at a and the tangent has the same equation.

So differentiability is *equivalent* to the tangent existing and being given by that expression — except in the case of a **vertical** tangent, which our definition of the derivative simply does not capture. (One could modify the definition to allow ∞, and then vertical lines would be covered too; we do not do that here.)

4 Computing tangent lines

Example 4.1

Find the tangent to $f(x) = 5x^3 - 17x^2 + \pi x - 0.5$ at $a = 0$.

We computed this derivative in the previous lecture:

$$f'(x) = 15x^2 - 34x + \pi \qquad \Longrightarrow \qquad f'(0) = \pi.$$

Also $f(0) = -0.5$. Therefore

$$y = f'(0)(x-0) + f(0) = \pi x - 0.5.$$

Example 4.2

Find the tangent to $f(x) = \cos x$ at $a = \dfrac{\pi}{3}$.

Here $f'(x) = -\sin x$, so

$$f'\left(\frac{\pi}{3}\right) = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}, \qquad f\left(\frac{\pi}{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2},$$

recalling that $\frac{\pi}{3}$ is $60°$. Hence

$$y = -\frac{\sqrt{3}}{2}\left(x - \frac{\pi}{3}\right) + \frac{1}{2}.$$

Example 4.3

Find the tangent to $f(x) = x\tan x$ at $a = \dfrac{\pi}{4}$.

This looks complicated, but it is a **product** of two functions whose derivatives we know, so apply the product rule with $\frac{d}{dx}(\tan x) = \sec^2 x$:

$$f'(x) = 1 \cdot \tan x + x \cdot \sec^2 x = \tan x + x\sec^2 x.$$

Now evaluate at $\frac{\pi}{4}$, that is at $45°$. Since $\tan\frac{\pi}{4} = 1$ and $\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, we get $\sec\frac{\pi}{4} = \sqrt{2}$ and hence $\sec^2\frac{\pi}{4} = 2$. So

$$f'\left(\frac{\pi}{4}\right) = 1 + \frac{\pi}{4}\times 2 = 1 + \frac{\pi}{2}, \qquad f\left(\frac{\pi}{4}\right) = \frac{\pi}{4}\tan\frac{\pi}{4} = \frac{\pi}{4}.$$

Therefore the tangent line is

$$y = \left(1 + \frac{\pi}{2}\right)\left(x - \frac{\pi}{4}\right) + \frac{\pi}{4}.$$

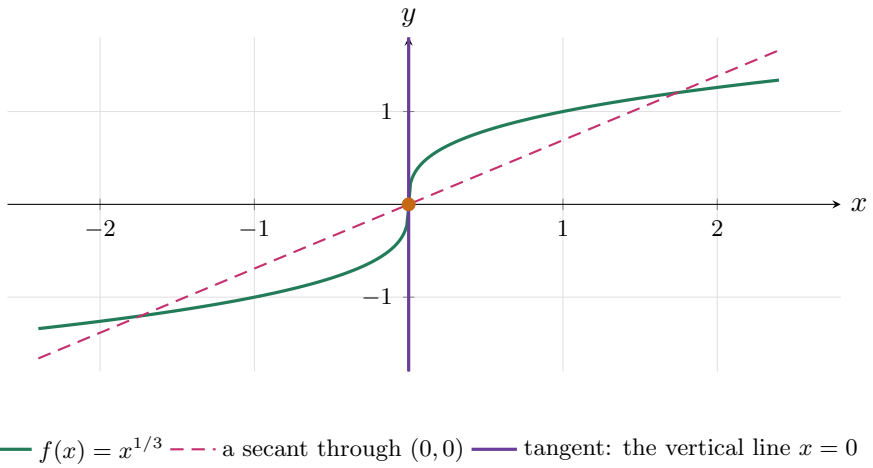
So once we know how to compute derivatives, we can actually write down the equation of the tangent line — something we can then plot on a graph.

5 A vertical tangent: $f(x) = x^{1/3}$

Example 4.4

What happens for $f(x) = x^{1/3}$ at $a = 0$?

Looking at the picture, the secants behave perfectly well: as the second point comes down to $(0, 0)$, they do approximate the tangent. And the instantaneous direction of the graph at $(0, 0)$ genuinely *is* vertical — the tangent is the y-axis, that is the line $x = 0$.



But now compute the derivative. For $x \neq 0$ the power rule gives

$$f'(x) = \frac{1}{3}\, x^{\frac{1}{3}-1} = \frac{1}{3}\, x^{-2/3} = \frac{1}{3\, x^{2/3}} = \frac{1}{3\sqrt[3]{x^2}},$$

— square x and take the cube root. At $x = 0$ there is plainly a problem, since x sits in the denominator. From first principles,

$$\lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h^{1/3} - 0}{h} = \lim_{h\to 0} \frac{1}{h^{2/3}},$$

which **diverges to infinity**, so in our notation the limit **does not exist** and f is **not differentiable at** 0.

The real issue

The tangent here is $x = 0$. Writing a line in the form $y = mx + c$ works only for lines that are **not vertical**, because m must be a real number. If we allowed $m = \infty$, we could divide through by m and obtain $\frac{1}{m} = 0$, giving $x = 0$ — salvaging the situation. That is exactly why Theorem 4.1 carries the caveat about the tangent not being vertical.

6 Linear approximation

Linear function

A **linear function** has the form $L(x) = c + dx$. It is convenient to write it as

$$L(x) = f(a) + m\,(x - a),$$

where m is the slope.

Why write it this way? Because putting $x = a$ gives $L(a) = f(a)$: the line passes through $\big(a, f(a)\big)$, so the graph of L meets the graph of f at that point. (With the same caveat as before: a vertical "linear function" would just be the line $x = a$.)

We want to choose the linear function that **best approximates** f near a:

$$f(x) \approx L(x) \qquad \text{for all } x \text{ close to } a.$$

What "best" means

This is a slightly technical notion. Among all possible linear functions, L is the best approximation if, for any *other* linear approximation, the difference $|f(x) - L(x)|$ is smaller than the difference between $f(x)$ and that other linear function, for all x in some small interval around a.

Theorem 4.2 — The best linear approximation

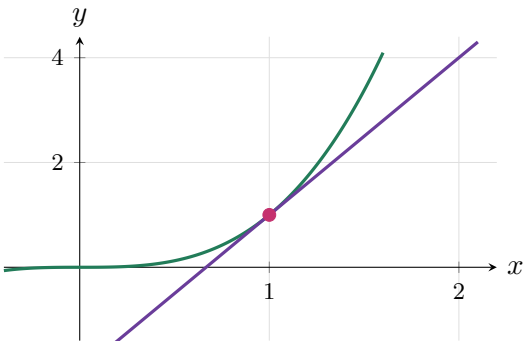
If f is differentiable at a, then the best linear approximation exists and is

$$\boxed{L(x) = f(a) + f'(a)\,(x - a)}$$

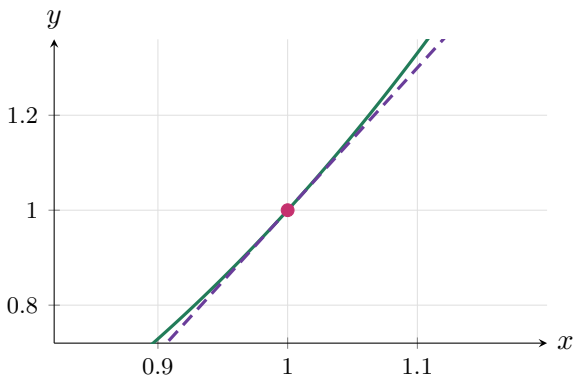
— exactly the expression for the **tangent**. Conversely, if a best linear approximation to f at a exists, then f is differentiable at a, and the approximation must be of this form.

This is intuitive from the pictures: as x comes closer to a, the graph of f is very closely approximated by the tangent — near the point of contact the two are almost indistinguishable. (A full proof is beyond this series; those mathematically inclined may attempt it from the definition of the tangent.)

away from $a = 1$ they differ



zoomed in: almost indistinguishable



6.1 Examples

Example 4.5

Find the best linear approximation to $f(x) = x^3$ at $a = 1$.

This is exactly the same computation as finding the tangent. Since $f'(x) = 3x^2$,

$$f'(1) = 3 \times 1^2 = 3, \qquad f(1) = 1,$$

so

$$L(x) = 3(x-1) + 1 = 3x - 3 + 1 = \boxed{3x - 2}$$

Example 4.6

Find the best linear approximation to $f(x) = \sec x$ at $a = 0$.

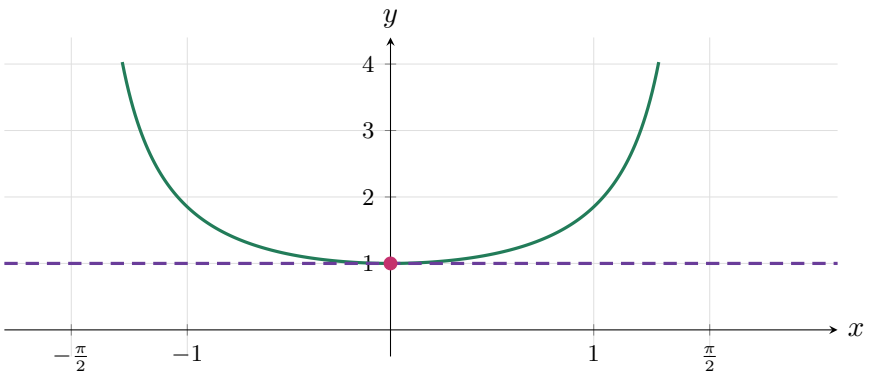
The derivative of $\sec x$ is $\sec x \tan x$, so

$$f'(0) = \sec 0 \cdot \tan 0 = 1 \times 0 = 0, \qquad f(0) = \sec 0 = 1.$$

Therefore

$$L(x) = 0 \cdot (x - 0) + 1 = \boxed{1}$$

— the **constant function** 1.



$f(x) = \sec x$ $L(x) = 1$ — a flat tangent

This is an interesting situation: the best linear approximation is a **constant function**, or equivalently the tangent is **flat**. That phenomenon is exactly what we take up next, under the name **turning points** — something you may recall from Mathematics–1.

The tangent at a has equation $y = f'(a)(x - a) + f(a)$ — and that same expression is the **best linear approximation** to f near a.

References and Acknowledgement

These notes are based on the video lectures of Mathematics–1 offered by the IIT Madras BS Degree Programme.

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